QUANT LAB USA INC

Internal Tools: Build vs Buy Decision Worksheet

A 20-page worksheet for ops teams evaluating Retool, Internal.io, Airplane, Tooljet, Appsmith, or building from scratch on Next.js plus Postgres. With a 10-factor scoring matrix, a 5-year TCO comparison, and an honest framework that does not pick a winner for you.

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About this document

This document is a free resource published by QUANT LAB USA INC, a Georgia-incorporated technology company building custom business software, SaaS platforms, and security engineering for mid-market and SMB operators. It is informational only and does not constitute legal, tax, accounting, or financial advice. Always verify decisions with a qualified licensed professional in your jurisdiction.

Section 1

Four-Question Filter

Most internal-tools decisions resolve on four questions. Answer these honestly before reading the platform comparison; you may not need it.

Question 1: Engineering Capacity

Do you have an engineering team that can build and maintain custom internal tools without sacrificing product velocity? If no, a no-code or low-code platform is almost certainly the right answer. If yes, the platform-vs-build trade-off becomes a financial and operational question rather than a capacity question.

Question 2: Data Sensitivity

How sensitive is the data the internal tool reads or writes? If it touches regulated data (HIPAA, PCI, financial-services regulated), the platform options narrow significantly. If it touches customer PII, the audit-logging maturity of the platform becomes critical. If it touches only operational data, most platforms are viable.

Question 3: Seat Count Trajectory

How many users do you need at launch and at 18 months? Per-seat platforms are cheap at small seat counts and expensive at large seat counts. The break-even point between Retool and a build-from-scratch alternative is typically 20-50 seats depending on tool complexity.

Question 4: Platform Standard Already?

Is your team already standardized on a particular platform? If yes, the switching cost of moving away is high. If no, you have full flexibility to pick on the merits.

Section 2

Platform Landscape

The major internal-tools platforms differ on pricing model, hosting, audit-logging maturity, and SDK surface. The table below summarizes the differences that matter most in vendor selection.

Platform	Pricing Model	Hosting	Best For
Retool	Per-seat	SaaS or self-hosted	Engineering-built admin panels
Internal.io	Per-seat	SaaS	Ops teams without engineering
Airplane	Per-task runs	SaaS or self-hosted	Scripted ops tasks
Tooljet	Open-source / hosted	Self-hosted	Cost-sensitive teams
Appsmith	Open-source / hosted	Self-hosted	Custom-skin internal apps
Budibase	Open-source / hosted	Self-hosted	Lightweight admin tools
Build-from-scratch	Eng time	Your stack	High customization needs

Retool

The market leader for engineering-built internal tools. Per-seat pricing scales predictably until you hit ~50 seats, at which point the bill becomes a meaningful line item. Best-in-class developer experience for SQL queries, API integrations, and data manipulation. Less suited for ops teams without engineering background; the tool assumes JavaScript fluency.

Internal.io

Lower-friction for non-engineering ops teams. Visual builder is more abstracted than Retool. SaaS-only - which is a constraint if data sensitivity rules out vendor-hosted data.

Airplane

Different model - tasks rather than apps. Best for scripted ops workflows that run on a schedule or on demand (run a SQL query, send a report, trigger a job). Less suited for interactive admin panels.

Open-source Options

Tooljet, Appsmith, and Budibase offer open-source self-hosted alternatives. The operational burden is higher (you manage the hosting, upgrades, and security). The cost trajectory is much flatter as seats grow. Best for cost-sensitive teams with operational capacity to host.

Section 3

Build-from-Scratch Baseline

A reference Next.js plus tRPC plus Postgres plus Auth.js stack handles most internal-tools requirements at first version cost of 6-10 engineering weeks. The ongoing maintenance cost is lower than the per-seat scaling cost of a platform at moderate-to-large seat counts.

Standard Stack

- **Frontend:** Next.js with App Router; shadcn/ui or Tailwind UI for shell.
- **Backend:** Next.js API routes or tRPC for type-safe RPCs.
- **Database:** Postgres (managed - Supabase, Neon, RDS, or similar).
- **Auth:** Auth.js (NextAuth) with SAML or OIDC integration.
- **Hosting:** Vercel for the app; managed Postgres separately.
- **RBAC:** Custom role table with middleware enforcement.
- **Audit logging:** Postgres audit table with hooks.

When Build-from-Scratch Beats Platform

Build wins when: customization depth is high; per-seat platform cost crosses the 50-seat break-even point; data sensitivity requires self-hosting; or integration depth into custom internal systems exceeds what a platform's SDK can express cleanly.

Section 4

10-Factor Scoring Matrix

Score each platform option (including build-from-scratch) on the 10 factors below, on a 1-10 scale. Apply weighting based on your situation.

Factor	Description	Default Weight
Seat count	How well does pricing scale to your seat count?	15%
Per-seat cost	Total cost at projected seat count.	10%
Integration depth	Can it integrate with your stack?	10%
Audit needs	SOC 2 / regulatory fit.	10%
Customization	How deep can you go before hitting limits?	10%
Data sensitivity	Self-hosted option available?	10%
Eng capacity	Required engineering time to operate.	10%
Vendor lock-in	Migration cost if you leave.	10%
Time-to-first-tool	How fast can you ship v1?	10%
Maintenance	Ongoing operational burden.	5%

Section 5

5-Year TCO Comparison

Side-by-side comparison of platform TCO versus build-from-scratch TCO across 5 years. Per-seat inflation modeled at 12% YoY. Maintenance modeled at 18% of build cost annually.

Year	Retool (50 seats)	Build-from-Scratch
Year 1	$24,000 + onboarding	$80,000 build cost
Year 2	$26,880	$14,400 maintenance
Year 3	$30,106	$16,128 maintenance
Year 4	$33,719	$18,063 maintenance
Year 5	$37,765	$20,231 maintenance
5-Year Total	$152,470	$148,822

At 50 seats, the 5-year TCO is roughly comparable. The break-even tilts toward build-from-scratch as seat count grows and toward platform as the engineering capacity to maintain a custom build shrinks. Re-run this math for your specific seat trajectory.

Section 6

Security and Access Control

Internal tools touch operational data and frequently customer data. The security posture of the tool matters more than is often recognized at procurement.

RBAC Maturity

Retool offers granular role-based access control across resources, queries, and components. Internal.io is comparable. Open-source platforms vary widely - Tooljet and Appsmith have improved RBAC significantly in recent releases but check against your specific requirements.

SSO and SAML

Single sign-on is table stakes for any company over ~50 employees. Verify SAML support is available on the tier you intend to purchase - several platforms gate SAML behind enterprise pricing tiers that materially shift the per-seat cost.

IP Allowlisting

If you require internal-tools access to be restricted to VPN or office IPs, verify the platform supports IP allowlisting at the tier you can afford. Some platforms gate this behind enterprise pricing.

Audit Log Retention

SOC 2 readiness reviews typically expect 12 months of audit log retention. Default retention varies by platform - confirm the retention period and any additional cost for longer retention.

Section 7

Audit and Compliance

If your business is in a regulated category (healthcare, financial services, insurance, legal services), the audit posture of your internal-tools stack matters for compliance reviews and customer attestations.

Data Residency

Confirm where the platform stores data - region and country. For EU-data scenarios, GDPR data-residency requirements may rule out US-only SaaS options.

PII Handling

Map which internal tool reads or writes customer PII. The customer-data exposure surface of an internal tool is part of your overall PII attack surface and figures into SOC 2 and customer security questionnaires.

Audit Trail Patterns

An auditor will want to see who accessed what data when. Verify the platform's audit log captures the events your auditor cares about - typically read events on sensitive tables, configuration changes, and access-control changes.

Section 8

Migration Risk

The most expensive failure mode in internal tools is committing to a platform, investing in 18 months of tool-building, and then needing to migrate. Understanding migration cost up front is part of vendor evaluation.

Retool-to-Custom Worked Example

A typical 30-tool Retool environment migrates to a custom build over approximately 16-20 engineering weeks. The rebuild cost is comparable to the original build from scratch (most of the cost is rebuilding the queries, components, and permissions; less of the cost is the underlying data model).

Mitigating Migration Risk

- Keep your data layer outside the platform - point Retool at your own Postgres, not at a Retool-hosted database.
- Document every tool's purpose and key queries in a separate registry.
- Avoid heavy use of platform-specific abstractions (Retool's transformers, Internal's custom code blocks) when standard SQL or REST patterns suffice.
- Standardize on patterns within the platform so a rebuild is mechanical, not exploratory.

Section 9

Hybrid Patterns

The most defensible internal-tools setups are often hybrid - a platform for simple tools, a custom build for complex ones.

Pattern 1: Platform for Reads, Custom for Writes

Use the platform for read-only dashboards, reporting, and operational monitoring. Use a custom build for write-heavy workflows (data correction, customer support actions, billing operations) where audit-logging and approval workflows matter.

Pattern 2: Platform for Generalist Tools, Custom for Specialist Workflows

Use the platform for tools used by 30+ ops users. Use a custom build for tools used by 3-5 specialists doing high-stakes operational work.

Pattern 3: Platform for Early-Stage, Custom for Scale

Start on a platform to get to v1 fast. Migrate to custom for the tools that hit scale-related issues - either seat count or customization depth. Treat the migration cost as a feature, not a bug.

Section 10

Decision Template

Use the one-page template below to summarize your scoring matrix output and present the decision to leadership. The template surfaces the three objections you will get and gives you responses.

One-Page Decision Template

- **Recommendation:** [Platform name] or [Build-from-scratch].
- **Scoring rationale:** [Top three reasons from the 10-factor matrix].
- **5-year TCO:** $[total] across [N] seats.
- **Switching cost if this is wrong:** $[migration cost].
- **Decision review milestone:** [Date when this decision is re-evaluated].

Three Objections and Responses

- **Objection 1: 'Why not just use Retool?'** - Response: [The matrix output for Retool versus the chosen path; show the math].
- **Objection 2: 'Why are we paying engineering for this?'** - Response: [The break-even math against the per-seat alternative at projected scale].
- **Objection 3: 'What if our needs change?'** - Response: [The migration scenario from Section 8 and the cost of changing direction in 18 months].

Contact & Next Steps

Ready to act on what is in this document?

If the contents of this PDF prompted a question you want a second set of eyes on, or if you are ready to scope a related engagement, we run a 20-minute discovery call at no cost. The conversation is structured: 5 minutes on your context, 10 minutes on the specific decision in front of you, 5 minutes on what good next steps look like.

- Book a 20-minute scoping call: **quantlabusa.dev/contact**
- Learn about our Custom Business Software practice: **quantlabusa.dev/services/custom-business-software**
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Who is QUANT LAB USA?

QUANT LAB USA INC is a Georgia-incorporated technology consultancy. We build custom business software, SaaS platforms, payments and licensing infrastructure, and security engineering for mid-market and SMB operators. We are deliberately structured as a small, senior-engineering-led practice - the people on your scoping call are the people who build the system. If a custom build is on the table, we typically begin with a paid discovery sprint that produces wireframes, a database schema, and a phased build estimate.

QUANT LAB USA INC
Georgia C-Corporation - SOS #26086454 - EIN 42-2039870
Web: quantlabusa.dev - Email: hello@quantlabusa.dev
Resources library: quantlabusa.dev/resources